

Conjecture 00000003423 is false:
adding an edge can increase hitting times

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1 The conjecture

Conjecture 00000003423 (English text, verbatim): “Definition: The maximum principle of hitting times: suprema on graphs. Conjecture: The supremum structure of hitting times: the maximal hitting time of the complete graph is $n - 1$, and adding edges decreases hitting time submodularly.” The Chinese text says the same.

Reading. The conjecture is a conjunction. Its first clause is true: on K_n every $H(a \rightarrow b)$, $a \neq b$, equals $n - 1$. We refute the second clause. “Adding edges decreases hitting time submodularly” contains the monotonicity claim: if $G \subseteq G'$ are connected graphs on the same vertex set (G' is G with edges added), then hitting times in G' are at most those in G ; submodularity is a further property of this decrease. We refute monotonicity (both weak and strict) in two readings of “hitting time”:

(R1) the pairwise hitting time $H_G(a \rightarrow b)$, for all pairs a, b ;

(R2) the maximal hitting time $\max_{a,b} H_G(a \rightarrow b)$, the quantity of the first clause.

Not covered: other graph parameters sometimes loosely called hitting times (commute times, averages of hitting times over a stationary start or target).

2 Definitions

Let G be a finite connected simple graph. The simple random walk $(X_t)_{t \geq 0}$ on G moves from v to a uniformly random neighbour: $P(v, u) = 1/\deg v$ for $u \sim v$. The hitting time of b is $\tau_b = \inf\{t \geq 0 : X_t = b\}$ (“the first time at which a given process hits a given subset of the state space”, Wikipedia, “Hitting time”), and $H_G(a \rightarrow b) = \mathbb{E}_a[\tau_b]$.

First-step analysis. $\tau_b = 0$ if $X_0 = b$; if $X_0 = v \neq b$, then τ_b is 1 plus the hitting time of b by the walk started at X_1 . By the Markov property $h(v) = \mathbb{E}_v[\tau_b]$ satisfies

$$h(b) = 0, \quad h(v) = 1 + \frac{1}{\deg v} \sum_{u \sim v} h(u) \quad (v \neq b). \quad (\text{S})$$

The values are finite on a finite connected graph: from every vertex, b is reached within $|V|$ steps with probability at least some $\delta > 0$, so τ_b has a geometric tail. Theorem 2 below shows that (S) has exactly one real solution. Hence $H_G(\cdot \rightarrow b)$ is the unique solution of (S), and this is how Lean defines it.

Definition 1 (as in Lean). `IsHittingTimeVec` $G \ b \ h$ is the system (S) with $\sum_{u \sim v}$ over Mathlib's `G.neighborFinset` v and $\deg v = G.degree \ v$. `hittingTime` $G \ a \ b$ is the value at a of a solution of (S), chosen by `Classical.epsilon`; by Theorem 2 this is the unique solution when G is connected. `maxHittingTime` $G = \max_{(a,b) \in V \times V} \text{hittingTime } G \ a \ b$.

3 Existence and uniqueness for (S)

Lemma 2 (maximum principle). *Let G be connected and $f : V \rightarrow \mathbb{R}$ with $f(b) = 0$ and $f(v) = \frac{1}{\deg v} \sum_{u \sim v} f(u)$ for $v \neq b$. Then $f \leq 0$, and hence $(-f)$ satisfies the same) $f = 0$.*

Proof. Let m maximise f and $M = f(m)$. If $x \neq b$, $f(x) = M$ and $x \sim y$, then $\deg x > 0$ and $\sum_{u \sim x} (M - f(u)) = \deg x \cdot M - \deg x \cdot f(x) = 0$ with nonnegative terms, so $f(y) = M$. Following a walk from m to b (it exists by connectedness), every vertex reached before b has value M , so $f(b) = M$, i.e. $M = 0$. $\square$

Theorem 3. *If G is connected, (S) has exactly one real solution.*

Proof. The difference of two solutions satisfies the hypothesis of the lemma, so it is 0. For existence, the linear map $L : \mathbb{R}^V \rightarrow \mathbb{R}^V$, $(Lh)(b) = h(b)$, $(Lh)(v) = h(v) - \frac{1}{\deg v} \sum_{u \sim v} h(u)$ ($v \neq b$), is injective by the lemma, hence surjective because $\mathbb{R}^V$ is finite-dimensional. A preimage of the vector that is 0 at b and 1 elsewhere solves (S). $\square$

4 The counterexample

Let P_5 be the path $0 - 1 - 2 - 3 - 4$ (Mathlib's `pathGraph 5`) and $P_5^+ = P_5 \cup \{0, 2\}$ the graph obtained by adding the single edge $\{0, 2\}$. Both are connected and $P_5 \leq P_5^+$.

Hitting times on P_5 . For target j and start i the solution of (S) is $H(i \rightarrow j) = j^2 - i^2$ for $i \leq j$ and $(4 - j)^2 - (4 - i)^2$ for $i \geq j$:

target \ start	0	1	2	3	4
0	0	7	12	15	16
1	1	0	5	8	9
2	4	3	0	3	4
3	9	8	5	0	1
4	16	15	12	7	0

Each row is checked against (S) directly (for example, target 4, vertex 2: $1 + \frac{15+7}{2} = 12$). So $H_{P_5}(0 \rightarrow 4) = 16$ and $\max_{a,b} H_{P_5}(a \rightarrow b) = 16$.

Hitting times on P_5^+ , target 4. The neighbour sets are $N(0) = \{1, 2\}$, $N(1) = \{0, 2\}$, $N(2) = \{0, 1, 3\}$, $N(3) = \{2, 4\}$. The vector $h = (18, 18, 16, 9, 0)$ solves (S): $1 + \frac{18+16}{2} = 18$ (vertices 0 and 1), $1 + \frac{18+18+9}{3} = 16$, $1 + \frac{16+0}{2} = 9$. So $H_{P_5^+}(0 \rightarrow 4) = 18$ and $\max_{a,b} H_{P_5^+}(a \rightarrow b) \geq 18$.

Theorem 4. *Adding the edge $\{0, 2\}$ to P_5 raises $H(0 \rightarrow 4)$ from 16 to 18 and the maximal hitting time from 16 to at least 18. Hence monotonicity fails in readings (R1) and (R2), and the second clause of the conjecture is false.*

Intuition: from 0 the walk on P_5 must step to 1 and then to 2; with the chord it can jump to 2, but from 2 it now returns to $\{0, 1\}$ with probability $2/3$ instead of $1/2$.

5 Lean formalization

File `lean/Conjecture3423/Basic.lean` (242 lines, only import `Mathlib`), namespace `C3423`. General part: `IsHittingTimeVec`, `harmonic_le_zero` (Lemma 1), `isHittingTimeVec_unique`, `exists_isHittingTimeVec` (Theorem 2), `hittingTime`, `hittingTime_eq`, `hittingTime_spec`, `maxHittingTime`. Instance part: `P5chord := pathGraph 5  $\sqcup$  SimpleGraph.edge 0 2`; `P5_connected` (from `Mathlib`’s `pathGraph_connected`), `P5chord_connected`, `nbr_P5`, `nbr_P5chord` (by decide on `Fin 5`), `hP5_spec`, `hChord_spec`, `hittingTime_P5`, `hittingTime_P5_0_4` ($= 16$), `hittingTime_P5chord_0_4` ($= 18$), `maxHittingTime_P5` ($= 16$), `maxHittingTime_P5chord_ge` (≥ 18), `adding_edge_increases_hittingTime`. Main theorems:

- `not_edge_monotone_hittingTime:` `not (forall (G G' : SimpleGraph (Fin 5)) [DecidableRel G.Adj] [DecidableRel G'.Adj], G.Connected -> G <= G' -> forall a b, hittingTime G' a b <= hittingTime G a b);`
- `not_edge_monotone_maxHittingTime:` `the same with maxHittingTime G' <= maxHittingTime G.`

`#print axioms` for both main theorems, `adding_edge_increases_hittingTime` and `hittingTime_spec` gives `[propext, Classical.choice, Quot.sound]`; no sorry, no native decide. Checked with Lean 4.33.1 and `Mathlib v4.33.1` (lake build). In Lean, H is defined through (S) (Theorem 2); the identification of (S) with $\mathbb{E}_a[\tau_b]$ is the first-step analysis above, which is not formalized.

6 Relation to the earlier submission

An earlier submission (PR #224, orionsheep) was closed. It used $P_3 \rightarrow K_3$, where adding the edge $\{1, 3\}$ raises $H(1 \rightarrow 2)$ from 1 to 2. The reviewer wrote: “the Lean theorems are $2 \cdot 2 = 2 + 2$ (twice), $2 \cdot 2 - 1 \neq 0$, $1 = 1$, $2 > 1$; graphs, random walks, and hitting times appear nowhere, so the P_3 -vs- K_3 counterexample is exclusively prose”. This submission defines graphs (`Mathlib SimpleGraph`), the simple random walk’s hitting-time system and the hitting time itself, and proves existence and uniqueness of the solution on every connected graph. It also replaces the example. In $P_3 \rightarrow K_3$ the maximal hitting time drops from 4 to 2, so that example refutes only reading (R1). The $P_5 \rightarrow P_5^+$ example refutes (R1) and (R2).

Source

Wikipedia, “Hitting time”, https://en.wikipedia.org/wiki/Hitting_time (definition quoted above).